

Where the constraint migrates in biotech: finite atoms and human donors that capital cannot replicate

Six calls on inputs that scale with physics or human arms, not with money: isotope purity, legacy radium, enriched stable targets, donated plasma, donated platelets, and the organ pool GLP-1 quietly drains.

The frame: when a system scales, the money does not stay where the headlines are. It moves to the one input that cannot scale with it. Our job is to name that input, and the date it starts to bite, before the market prices it in.

Area: biotech and human health (drug modalities and delivery, biomanufacturing and fill-finish capacity, donor-derived and cell/gene inputs, diagnostics and tools/reagents, medical isotopes and devices, longevity and aging)

- Horizon: 2030 to 2040

How we work: cast wide for ideas, then test every one hard against the price and the supply chain. We name the specific input, not the theme.

Two numbers per call: how strong we think the case is, and the odds on the exact, dated version, which is the one we are graded on.

Status: the calls that survived our own attempt to break them. Drafted 2026-06-14.

The board: 6 calls that survived the cut

The cross-cutting read

We make the same structural argument six times. In each case a loud, well-funded boom is paying for the visible layer (curie count, fractionation plants, targeting ligands, reconditioning devices) while the binding constraint sits one layer upstream, on an input that physics or biology will not let anyone manufacture on the relevant timescale. Three of these are medical-isotope calls, and we want to be clear they are not the same call: they name three genuinely different inelastic inputs. The Ac-227 call turns on isotopic inseparability. Ac-225 and Ac-227 are the same element in the same oxidation state, so every scaled accelerator route co-produces a radiotoxic impurity that no chemistry removes, and only the finite Th-229/U-233 generator stock is clean by construction. The Ra-226 call turns on quantity. Roughly 2.5 kg of legacy radium was ever extracted, the IAEA "fix" is a bounded gleaning of antique hospital sources rather than a manufacturing ramp, and building more accelerators only intensifies competition for that fixed stock. The stable-isotope call turns on geography. Kilogram-scale gas-centrifuge enrichment of Yb-176, Zn-68 and Gd-160 (the targets that breed Lu-177, Cu-67, Tb-161) sits overwhelmingly in Rosatom cascades, and Western EMIS and laser rebuilds remain gram-scale and isotope-narrow. The two donor-derived calls share the deepest physical truth. Polyclonal IgG and platelet-lysate growth factors are not molecules anyone synthesizes, they are pooled human repertoires whose therapeutic property is donor diversity, so supply is hard-bounded by eligible donors times a regulated maximum donation rate. The GLP-1 call flips the consensus sign. By removing the metabolic and cardiovascular deaths that feed the marginal donor stream, GLP-1 degrades rather than improves the usable organ pool. The seam is consistent across all six, and we hold the dated-call probabilities honestly below our conviction in the theses for one recurring reason: these constraints clear through government allocation tables, bilateral contracts, and informal hospital rationing rather than through Bloomberg-trackable spot prices, so the structural call can be right while the dated, publicly-cited resolution criterion fails to print on schedule.

At a glance

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
P1	Targeted alpha therapy is the boom: Ac-225-PSMA for prostate cancer, Ac-225-DOTATATE for neuroendocrine tumors, and a wide wave of antibody- and small-molecule-targeted alpha conjugates. Analysts model 20 to 44 percent CAGR toward 1 to 4 billion dollars by 2030, with at least nine commercial alpha products expected by 2030,	Isotopically pure, Ac-227-free Ac-225 sourced from the finite, non-replenishing Th-229/U-233 stockpile and the handful o	74%	57%	2032-12-31

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
	and the pharma majors are racing to lock supply (Lilly/Point, Novartis, BMS/RayzeBio, AstraZeneca, Bayer). The prevailing read treats raw Ac-225 curie availability as the bottleneck and the targeting vector as the place innovation happens.				
P2	An Ac-225-based targeted alpha therapy reaches blockbuster volume. The field treats Ac-225 atoms and accelerator beam-time as the things to scale, while the non-replenishable Ra-226 target feedstock quietly becomes the true ceiling.	Purified Ra-226 target feedstock (roughly 2.5 kg total global legacy stock, non-manufacturable, recoverable only from an	78%	44%	2034-12-31
P3	Pluvicto and Lutathera turned beta-emitting radioligand therapy into a multi-billion-dollar oncology category, and capital is now flooding the headline layer: targeting ligands, reactor slots, cyclotrons, and the radioisotopes themselves (Tb-161, Cu-67, Ac-225). The reactor chokepoint is the public narrative.	Kilogram-scale gas-centrifuge enrichment of stable-isotope targets (Yb-176, Zn-68, Gd-160) concentrated in Rosatom plant	72%	38%	2032-12-31
P4	Polyclonal IgG is the pooled antibody repertoire of thousands of donors. Unlike any monoclonal or recombinant biologic, you cannot grow it in a cell line, because the therapeutic property is the diversity of the pooled-donor immune repertoire rather than a single sequence. That biological fact is permanent and non-negotiable. Three	Human source-plasma collection litres from the US-concentrated, compensated donor base -- the irreplaceable polyclonal-I	82%	40%	2035-12-31

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
	compounding forces pull demand upward: aging populations with primary immunodeficiency, expanding approved neurology indications (CIDP, MG, multifocal motor neuropathy), and a growing population of cancer-immunotherapy survivors with secondary immunodeficiency. IG demand is tracking roughly 30 percent growth by 2030 and rising into the 2030s. Supply is litres of source plasma, 70 percent of it collected from US compensated donors under regulatory and physiological frequency caps that set a hard ceiling on litres per donor per year. Each lot needs 7-12 months from collection to finished vial. The system cannot sprint. The rent concentrates in whoever controls donor-centre litres, and that input cannot be financialized, engineered, or scaled the way a factory can.				
P5	The allogeneic off-the-shelf cell therapy and regenerative-medicine wave: MSC products, iPSC-derived therapies, allogeneic CAR-T and NK products, and the broader stem-cell pipeline that has to expand cells to billions of doses in bioreactors. Regulatory and quality pressure is at the same time forcing the field off FBS, which is itself supply-capped and carries zoonotic risk, onto a human-derived substitute whose supply is biologically	GMP-grade human platelet lysate derived from pooled volunteer-donor blood-bank platelet units. The inelastic input is hu	72%	38%	2033-12-31

#	THE BOOM	BINDING CONSTRAINT (THE NEEDLE)	CASE	OUR CALL	RESOLVES
	constrained and whose donor pool is already strained for ordinary clinical transfusion.				
P6	The loud GLP-1 narrative is about supply and demand in obesity/diabetes therapeutics and downstream food/device/CDMO effects. The transplant-adjacent GLP-1 conversation is entirely about GLP-1 as therapy for recipients and for reversing donor-organ steatosis. Everyone treats GLP-1 as straightforwardly good for organ health. Nobody models GLP-1 as a structural drag on the donor pool itself.	Standard-criteria-donor (SCD) organs: young, low-comorbidity deceased-donor solid organs usable without reconditioning.	68%	38%	2033-12-31

Case = how strong we think the thesis is. Our call = the probability we put on the exact, dated version landing by the deadline, and the number we are graded on. The gap between them is the cost of pinning a date and a hard threshold, not a lack of conviction.

P1 • Isotopically pure (Ac-227-free) Ac-225 from the finite Th-229/U-233 stockpile is the physics-locked binding constraint on targeted-alpha therapy through the 2030s, not raw curie count

P1

The boom: Targeted alpha therapy is the boom: Ac-225-PSMA for prostate cancer, Ac-225-DOTATATE for neuroendocrine tumors, and a wide wave of antibody- and small-molecule-targeted alpha conjugates. Analysts model 20 to 44 percent CAGR toward 1 to 4 billion dollars by 2030, with at least nine commercial alpha products expected by 2030, and the pharma majors are racing to lock supply (Lilly/Point, Novartis, BMS/RayzeBio, AstraZeneca, Bayer). The prevailing read treats raw Ac-225 curie availability as the bottleneck and the targeting vector as the place innovation happens. • Domain: biotech and human health (medical isotopes / radiopharmaceuticals)

STRUCTURAL CASE

74%

OUR CALL, DATED

57%

RESOLVES

2032-12-31

Alpha emitters kill tumors by delivering high-LET, double-strand DNA breaks over a few cell diameters. Nothing chemical or biological substitutes for that: the physics of the decay is the drug. Historically all clinical-grade Ac-225 came from Th-229 generators isolated from an aged U-233 stockpile built for weapons and reactor programs. Th-229 decays to Ac-225 with no Ac-227 branch, so generator-derived material is isotopically clean. Legacy U.S. supply at ORNL runs on the order of 1 Curie per year. U-233 production is permanently shut down under current policy, which makes this source finite and non-replenishing. Every scaled accelerator route now in development (proton or deuteron irradiation of Ra-226 targets, spallation of Th targets, photonuclear routes targeting more than 100 Ci/yr by 2029) co-produces Ac-227 because the nuclear reactions sit right next to each other. Since Ac-225 and Ac-227 are the same element in the same oxidation state, no chemical separation is possible. Ac-227 carries a 21.8-year half-life, so it adds a persistent radiotoxic burden to the patient and the waste stream. The Ac-227 ceiling on cumulative patient dose tightens as TAT regimens move from single doses to repeat or chronic administration, and emerging indications are pushing exactly that way. NorthStar's April 2026 FDA DMF acceptance for "no-carrier-added" Ac-225 from a cyclotron route shows the industry treats purity as a quality attribute, but it has not yet resolved, in public regulatory language or sell-side models, the gap between carrier-free chemistry purity and Ac-227 isotopic purity. RayzeBio's Phase 3 pause over an Ac-225 shortage is the early tremor that supply gating is real; the purity dimension is the next layer, surfacing as curies scale but clean curies do not.

WHY IT IS NOT PRICED YET

The market has loudly noticed raw Ac-225 scarcity and is pouring capital into curie expansion (Cardinal, NorthStar, TerraPower, PanTera, SHINE, Eckert and Ziegler). That first-order shortage already sits in pharma supply agreements and equity models. What does not sit there is the second-order, physics-locked point: scaled routes co-produce Ac-227 inseparably, so adding curies does not add clean curies. Sell-side models treat Ac-225 supply as a single fungible number, never separating generator-grade from accelerator-grade and never pricing the Ac-227 ceiling on cumulative patient dose. NorthStar's April 2026 DMF acceptance leans on "no-carrier-added" language that conflates chemistry purity with isotopic purity, a sign that even regulators and suppliers have not yet formally split the two. The purity premium is forming, but it has not yet shown up in contract pricing or drug master file specifications in the public record. We read this as a genuine

seam that the existing FUTURE_MAP biotech calls do not cover, since none of them touch medical isotope chokepoints.

WHERE THE PRICE SITS TODAY

Not priced at the isotopic-purity layer. Raw Ac-225 scarcity shows up in pharma supply agreements and private-market valuations for isotope producers. The Ac-227 purity distinction does not yet appear in public sell-side models, contract pricing disclosures, or regulatory guidance separating generator-grade from accelerator-grade for repeat-dosing indications. NorthStar's NCA DMF language is the closest public signal, and it does not resolve the Ac-227 versus carrier-free ambiguity. The price channel is partially obscure on the needle as specified.

THE BINDING CONSTRAINT

Isotopically pure, Ac-227-free Ac-225 sourced from the finite, non-replenishing Th-229/U-233 stockpile and the handful of generators derived from it. The inelastic input is not gross curie output but isotopic purity: the absence of the chemically inseparable Ac-227 co-impurity that physics dictates every scaled accelerator and reactor route introduces.

WHAT WE ARE WATCHING

Two tracked series, both annual from 2026. (1) Curies per year of isotopically pure, generator-sourced (Th-229/U-233) Ac-225 with certified Ac-227 content below the dose-acceptable threshold, versus total Ac-225 curies produced including accelerator/Ra-226/photonuclear routes. Current generator-grade supply is approximately 1 Ci/yr at ORNL while accelerator targets aim for 100-plus Ci/yr by 2029. (2) The price and contract-allocation spread per millicurie between certified-low-Ac-227 lots and accelerator-route lots, plus the count of TAT clinical or commercial programs that specify a maximum Ac-227 impurity concentration in their drug master file or supply contract. Resolution criterion: does the clean-grade share remain a small minority of total supply while commanding a rising purity premium, and does Ac-227 impurity become an explicitly cited gating specification in TAT supply agreements by 2032?

WHAT WOULD PROVE US WRONG

Kill if by 2032 an accelerator/reactor/photonuclear route reaches routine multi-tens-of-Curie annual output of Ac-225 whose Ac-227 content is certified low enough that regulators and sponsors treat it as interchangeable with generator-grade for chronic or repeat dosing (purity premium collapses to under 15 percent and no major TAT program cites Ac-227 as a supply or dosing constraint). Also kill if practical electromagnetic or mass-selective isotope separation removes the Ac-227 impurity at commercial scale, or if the field migrates decisively to Pb-212 or Tb-149 alpha emitters that sidestep the Ac-225 purity problem. Partial kill: if U-233 production is reinstated under a policy reversal, the finite-stockpile premise weakens even if the purity advantage of Th-229-derived material stays intact.

HOW WE TRIED TO BREAK IT

Three attacks survive, at reduced severity. First, single-dose or low-repeat TAT protocols (the current standard) may never accumulate Ac-227 to a regulatory ceiling, which keeps the purity constraint theoretical unless chronic dosing scales. That is the load-bearing assumption. Second, U-233 production closure is policy, not physics, so a future nuclear-materials program could reopen the route and weaken the finite-stockpile premise. Third, Pb-212 alpha therapy (from Ra-228/Th-228 generators) is a live alternative modality that sidesteps the Ac-227 problem entirely, and if it captures a large share of the TAT pipeline the purity

constraint goes niche. None of these kills the call today, but together they set the real odds below the structural ceiling because each is plausible inside the resolution window.

Why we are making the call The physics of co-production is locked. Ac-225 and Ac-227 are the same element, and every accelerator scaling route makes both. The Th-229/U-233 source is the only route that avoids this by construction, and that source is permanently capped under current policy. The purity premium is not yet priced into equity models or supply contracts. The call survives all three adversarial attacks at the structural level. We hold the dated-call probability below our conviction in the thesis because the resolution criterion needs observable market signals (purity premiums in contracts, Ac-227 specs in DMFs) that are forming but not yet locked, and because the chronic-dosing proliferation timeline stays uncertain inside the 2026 to 2032 window.

P2 · By 2034, the binding constraint on the Ac-225 targeted-alpha-therapy boom is not the accelerator conversion step but the finite, non-manufacturable Ra-226 target feedstock that every accelerator route consumes, and the hot-cell capacity to fabricate, irradiate, and recover it.

The boom: An Ac-225-based targeted alpha therapy reaches blockbuster volume. The field treats Ac-225 atoms and accelerator beam-time as the things to scale, while the non-replenishable Ra-226 target feedstock quietly becomes the true ceiling. · Domain: biotech

STRUCTURAL CASE

78%

OUR CALL, DATED

44%

RESOLVES

2034-12-31

Ac-225 production runs through three serial layers: Ra-226 source, accelerator conversion, and GMP radiochemistry. The funded buildout (Niowave, Actineer/CNL cyclotron already running in 2025, DOE photonuclear programs) attacks layer 2. But Ra-226 is a secular decay product of uranium, never made on purpose. Roughly 2.5 kg was extracted worldwide across the 20th century, and the only new supply is scavenging antique radium medical sources under the IAEA global radium-management initiative. Accelerator routes do not consume Ra-226 atoms permanently if the target is recycled, but imperfect recovery plus limited hot-cell fabrication capacity means the effective Ra-226 inventory shrinks with throughput. As conversion capacity gets funded and built, the binding constraint walks upstream, deterministically, to the fixed Ra-226 stock and the shielded hot-cell capacity to work that stock without loss. This rhymes with the He-3 case: a tiny, byproduct-only legacy inventory that capital cannot replicate on a decade timescale.

WHY IT IS NOT PRICED YET

The investor and trade narrative has settled on "Ac-225 is scarce, fund accelerators and hot-cells." IAEA, DOE/NIDC, and the trade press (AuntMinnie, Clinical Trial Vanguard) already name Ra-226 sourcing as a feedstock concern, so the framing is not buried. It has not, though, crossed into sell-side equity models, where accelerator capacity remains the consensus binding variable. The second-order migration (solving conversion just exposes the Ra-226 floor) is absent from financial modeling. We think the mispricing is real but eroding, since awareness keeps rising in technical and institutional channels. The cleanest tell that financial markets have not priced this yet is the price-channel leg: there is no tracked spot price for Ra-226.

WHERE THE PRICE SITS TODAY

No public spot or contract price for purified Ra-226 exists. DOE/NIDC sells research quantities under administrative allocation, not open-market bidding. So the structural constraint is real, but price discovery may stay inside government allocation tables rather than a Bloomberg-trackable index, which is the primary risk to the dated call resolving regardless of whether the structural call is correct.

THE BINDING CONSTRAINT

Purified Ra-226 target feedstock (roughly 2.5 kg total global legacy stock, non-manufacturable, recoverable only from antique radium sources) plus the GMP shielded hot-cell capacity to fabricate, irradiate, and chemically recover those targets without radium loss.

WHAT WE ARE WATCHING

Public citations by an Ac-225 sponsor or CDMO of Ra-226 target availability as the rate-limiting input; emergence of a tracked spot or contract price for purified Ra-226 (\$/mg or \$/Ci); reported Ra-226 inventory recovered under the IAEA radium initiative versus annual Ac-225 demand. Baseline: global Ac-225 roughly 1.7 Ci/yr (enough for about 2000 patients); no public Ra-226 price index; DOE/NIDC lists Ra-226 as available in research quantities only; accelerator-conversion buildout is the entire headline story in equity and trade press.

WHAT WOULD PROVE US WRONG

The call is dead if, by 2034, accelerator buildout makes Ac-225 broadly available with no sponsor or CDMO citing Ra-226 target feedstock or target-fab hot-cell capacity as rate-limiting and no Ra-226 price or inventory tension emerges; OR if a Ra-226-free route to Ac-225 (high-energy proton spallation of Th-232 at TRIUMF or DOE scale, or Ra-226-free reactor routes) scales enough to make Ra-226 inventory irrelevant to supply; OR if Ac-225 alpha therapies fail pivotal efficacy or toxicity readouts and Lu-177 beta therapy remains dominant at scale.

HOW WE TRIED TO BREAK IT

Three adversarial attacks. (1) Ra-226-free routes. High-energy proton spallation of Th-232 at TRIUMF and BNL produces Ac-225 without Ra-226 targets. If these scale commercially before 2034, Ra-226 becomes irrelevant. The risk is real, but multi-GeV proton accelerators are expensive and sparse, and commercial displacement of Ra-226 routes by 2034 is not current consensus. (2) Already named publicly. NIDC, IAEA, and AuntMinnie already cite Ra-226 sourcing. If a sell-side analyst picks this up soon, the "not priced" premise weakens fast. The resolution test asks for a tracked spot price, which needs open-market price discovery, and government allocation could block that even as the constraint bites. (3) Target recycling loop. Accelerator routes can recycle Ra-226 after irradiation if hot-cell recovery is efficient. The constraint then shifts to recovery yield and hot-cell capacity rather than Ra-226 atoms outright. The needle should name both Ra-226 inventory and hot-cell recovery capacity together, which the candidate already does. The structural logic survives all three, but the exact dated call stays vulnerable to the government-allocation pricing mechanism.

Why we are making the call We promote on structural strength: the Ra-226 supply inelasticity is physically real, the constraint-migration logic from conversion to feedstock is deterministic as accelerator capacity builds, and the He-3 comparison holds. The mispricing is genuine in financial channels. We hold the dated-call probability to 0.44 because the resolution test needs a tracked spot or contract price for Ra-226, which may never emerge if DOE keeps allocating administratively rather than via open-market bidding, and because Ra-226-free spallation routes are a live kill-line risk. The structural view (Ra-226 becomes the real ceiling) has a high probability of being correct; the dated call is tighter than the structural story alone.

P3 · The binding constraint on the RLT boom migrates off the reactor onto kilogram-scale enriched stable-isotope feedstock (Yb-176, Zn-68, Gd-160) concentrated in Russian gas-centrifuge cascades, with Western rebuilds too gram-scale and too isotope-narrow to close the gap across the full therapeutic basket by 2032.

P3

The boom: Pluvicto and Lutathera turned beta-emitting radioligand therapy into a multi-billion-dollar oncology category, and capital is now flooding the headline layer: targeting ligands, reactor slots, cyclotrons, and the radioisotopes themselves (Tb-161, Cu-67, Ac-225). The reactor chokepoint is the public narrative. · Domain: biotech-health / medical isotopes / radiopharmaceuticals / stable-isotope enrichment

STRUCTURAL CASE

72%

OUR CALL, DATED

38%

RESOLVES

2032-12-31

A therapeutic radioisotope is bred, not mined. No-carrier-added Lu-177 needs an ytterbium-176-enriched target bombarded in a high-flux reactor; Cu-67 needs zinc-68; Tb-161 needs gadolinium-160. Enriching a stable isotope past 98 percent at kilogram scale takes gas-centrifuge cascades, and that capacity sits overwhelmingly at Rosatom's Electrochemical Plant (Zelenogorsk) and Ural Electrochemical (Novouralsk). The reactor or cyclotron only activates the target; it cannot produce the feedstock. Western centrifuge capacity for stable isotopes was retired in the 1990s. The rebuild is underway via EMIS (Kinectrics, ORNL SIPRC) and laser enrichment (ASP Isotopes ASPI, Pretoria), but Kinectrics EMIS tops out near 500 g/yr for Yb-176 and ASP Isotopes has not disclosed sustained kilogram-per-year throughput at clinical grade. Zn-68 and Gd-160 have no announced kilogram-scale Western rebuild programs. As RLT scales 5-10x and diversifies across isotopes, the rent step-shifts to whoever controls enriched-stable-target supply, one layer above the reactor, and the West has identified the dependency without solving it at basket scale. The constraint is isotope-specific and slow to qualify under GMP, so it stays genuinely inelastic through 2030 even though capital is now allocated.

WHY IT IS NOT PRICED YET

The first-order dependency (Yb-176 from Russia for Lu-177) now shows up in trade press, EU consortium papers (EURASIS 2024), and equity narratives (ASPI listed on Nasdaq explicitly to trade this thesis). That layer is partially priced. The second-order claim is less priced: the constraint is not the reactor, not Lu-177 specifically, and not Yb-176 alone, but the upstream gas-centrifuge enrichment step shared across the entire next generation of therapeutic isotopes as a basket, with Western rebuilds solving one isotope partially while leaving Zn-68 and Gd-160 unaddressed, and with allocation contracts keeping the bottleneck invisible to spot-market signals and equity coverage. Policy and equity narratives price the drug and the reactor while overlooking the centrifuge two layers upstream. Western EMIS announcements (Kinectrics 500 g/yr) get treated as a solution when the demand arithmetic does not support that at full basket scale. This is distinct from the existing biotech calls in FUTURE_MAP (delivery, AAV, plasmid, ADC conjugation), none of which touch medical isotopes or stable-isotope enrichment.

WHERE THE PRICE SITS TODAY

Partially priced at the first-order level via ASPI equity and DOE SIPRC funding. The second-order basket argument (Zn-68, Gd-160 remaining unaddressed; enriched-target allocation as the shared constraint across all next-generation therapeutic isotopes) is not reflected in any visible spot price or dedicated equity vehicle. Enriched target prices get set by bilateral allocation contracts without public disclosure, which keeps the constraint structurally invisible to markets even when it is operationally real.

THE BINDING CONSTRAINT

Kilogram-scale gas-centrifuge enrichment of stable-isotope targets (Yb-176, Zn-68, Gd-160) concentrated in Rosatom plants; Western EMIS and laser rebuilds are gram-scale for Yb-176 and nonexistent at scale for the rest of the therapeutic basket.

WHAT WE ARE WATCHING

Share of Western RLT therapeutic-isotope feedstock (enriched Yb-176, Zn-68, Gd-160) sourced from non-Russian kilogram-scale enrichment, plus disclosed Yb-176 unit price and lead time. Mid-2026 baseline: NCA Lu-177 supply chain still Russia-dependent for Yb-176 at commercial scale; Kinectrics targets 500 g/yr Yb-176; ASP Isotopes has produced commercial samples but no disclosed kg/yr throughput; no Western kilogram program announced for Zn-68 or Gd-160; enriched-target lead times measured in months and allocated by quiet bilateral contract rather than spot market.

WHAT WOULD PROVE US WRONG

Western or allied centrifuge-equivalent stable-isotope enrichment reaches qualified kilogram-scale output for at least two of {Yb-176, Zn-68, Gd-160} before 2030 (e.g. ASP Isotopes MLIS scales to greater than 2 kg/yr at clinical grade for Yb-176 AND a second target isotope enters a kilogram-scale Western program), OR accelerator-direct production routes that bypass enriched stable targets scale enough to supply commercial RLT without target enrichment, OR RLT demand growth stalls such that enriched-target allocation pressure is not rate-limiting for any program.

HOW WE TRIED TO BREAK IT

Three attacks survive partial scrutiny. First, the resolution test requires public citation by RLT sponsors of enriched-target supply as the rate-limiting input over reactor time. CDMOs and sponsors routinely describe supply chain risk in vague 10-K language, and if allocation is secured bilaterally, sponsors have no incentive to disclose the specific bottleneck, which makes the evidentiary bar hard to meet even when the bottleneck is real. Second, the Kinectrics plus ASPI plus SIPRC overlap could solve Yb-176 specifically by 2029, splitting the dated call: Western capacity may exceed demand for Lu-177 while staying inadequate for Tb-161 and Cu-67, producing an ambiguous resolve rather than a clean TRUE. Third, six years is a long runway. The West has identified the dependency and capital is moving, so by 2032 the basket may be partially addressed for the leading commercial isotopes even if not fully closed. The structural observation survives all three attacks; the specific dated call is the weaker component.

Why we are making the call We promote because the second-order structural mechanism is correct and inelastic through the mid-term, the basket argument (Zn-68, Gd-160 have no Western kilogram rebuild) extends past what equity or policy has priced, and the allocation-contract invisibility is a genuine market-structure reason the bottleneck does not self-disclose. We hold the dated-call probability below 0.5 because the evidentiary bar (public citation by sponsors of enriched target as

rate-limiting input over reactor time, for two isotopes) may not be met on schedule even when the bottleneck is operationally real, and because ASP Isotopes plus Kinectrics could solve Yb-176 before 2032 and split the resolve.

P4 · By 2035 the binding constraint on IVIG/SCIG supply is the human source-plasma donor base, a demographically and geographically concentrated input with no recombinant substitute; structural rationing or allocation tightens in at least two major wealthy markets, and the US share of world plasma collection stays above ~65 percent even as global IG demand grows materially.

The boom: Polyclonal IgG is the pooled antibody repertoire of thousands of donors. Unlike any monoclonal or recombinant biologic, you cannot grow it in a cell line, because the therapeutic property is the diversity of the pooled-donor immune repertoire rather than a single sequence. That biological fact is permanent and non-negotiable. Three compounding forces pull demand upward: aging populations with primary immunodeficiency, expanding approved neurology indications (CIDP, MG, multifocal motor neuropathy), and a growing population of cancer-immunotherapy survivors with secondary immunodeficiency. IG demand is tracking roughly 30 percent growth by 2030 and rising into the 2030s. Supply is litres of source plasma, 70 percent of it collected from US compensated donors under regulatory and physiological frequency caps that set a hard ceiling on litres per donor per year. Each lot needs 7-12 months from collection to finished vial. The system cannot sprint. The rent concentrates in whoever controls donor-centre litres, and that input cannot be financialized, engineered, or scaled the way a factory can. · Domain: biotech-health / donor-derived biologics (plasma fractionation)

STRUCTURAL CASE

82%

OUR CALL, DATED

40%

RESOLVES

2035-12-31

Source plasma comes from humans at a near-fixed litres-per-donor-per-year ceiling set by regulation and physiology. The donor pool concentrates in one country and skews toward a compensated demographic. There is no recombinant escape for polyclonal IgG because the therapeutic property IS the pooled-donor repertoire diversity. Demand runs on aging plus neurology indication expansion plus immunocompromised-survivor population growth; supply runs on human arms and a 7-12 month fractionation pipeline. The curves are structurally decoupled, and the input is non-substitutable and demographically capped.

WHY IT IS NOT PRICED YET

Partially mispriced. The biology of non-substitutability is a permanent fact and nobody contests it. The shortage risk, though, is more priced than the candidate implies. Grifols, CSL Behring, and Takeda are all publicly traded or carry public debt; sell-side analysts at Jefferies, Morgan Stanley, and Berenberg have covered plasma collection volume as the primary operating driver for at least five years. The "US supplies ~70 percent of world plasma" figure shows up in WHO working group documents and regulatory filings. What stays genuinely underpriced: the convergence of three demand curves at once (neurology expansion, immunocompromised-survivor population, aging-primary-immunodeficiency), the specific demographic fragility of compensated donation under any economic-normalisation scenario, and the absence of any FcRn-class substitute for the primary-immunodeficiency (non-neurology) IVIG base. The claim that this is entirely un-modeled is too strong; the claim that the full binding severity is priced is also too strong. The truth sits in between, which is why we promote but hold the dated-call probability well below our conviction in the thesis.

WHERE THE PRICE SITS TODAY

Plasma fractionator equities trade on collection-volume growth as the primary lever; the shortage risk is a standing topic on earnings calls. Market research projections (USD 13B to USD 23B by 2035, CAGR ~5.9 percent) get cited widely. FcRn antagonist launches (efgartigimod, rozanolixizumab) are already partially priced as a neurology-demand offset. The US-geographic concentration sits in public regulatory filings. This is not a dark, uncovered corner of the market.

THE BINDING CONSTRAINT

Human source-plasma collection litres from the US-concentrated, compensated donor base -- the irreplaceable polyclonal-IgG feedstock. Not fractionation plant capacity, not albumin or Factor co-products, not any recombinant alternative (which cannot exist for polyclonal IgG).

WHAT WE ARE WATCHING

US share of global plasma collection (currently near 70 percent); IG demand growth vs collected source-plasma litre growth (demand tracking roughly +30 percent by 2030, collection per capita roughly flat); number of major wealthy markets (US, EU member states, Canada, Japan) imposing IVIG allocation, prioritization criteria, or formal rationing. Current state: chronic localized IVIG shortages recurring, non-self-sufficient nations dependent on US plasma, a 2025 PMC study documenting 9 percent therapy-delay rates from shortages, and donor management flagged as a demographic risk in regulatory filings.

WHAT WOULD PROVE US WRONG

By 2035 either (a) FcRn antagonists (efgartigimod class) or other recombinant immunomodulators substitute for enough IVIG demand in the neurology indications that pooled-plasma IgG stops being the binding modality for the fastest-growing demand cohort, OR (b) plasma collection broadens enough via EU/Asia self-sufficiency programs or higher per-donor yield that collected litres outpace IG demand, US share of collection falls below 60 percent, and no major market resorts to IVIG allocation. Either dissolves the donor-base constraint as the binding needle.

HOW WE TRIED TO BREAK IT

Three attacks. First, FcRn antagonists are not hypothetical: argenx's efgartigimod (Vyvgart) is commercially approved for MG and CIDP and growing, competing directly for the highest-dose IVIG users in neurology, the exact demand cohort the candidate calls open-ended. Over a 9-year horizon, this class is a material kill path, not a speculative one. Second, EU plasma self-sufficiency is an active policy target: the European Commission's Plasma Action Plan (2023) and post-COVID health-security legislation have made non-US plasma collection a genuine political priority with capital behind it, and nine years is enough time to move the US share below 65 percent. Third, the rationing sub-condition is fragile: formal public rationing in two major wealthy markets requires both governments and hospital systems to move from informal shortage management (the current state) to declared allocation frameworks, which has not happened despite years of recurring shortages, suggesting the system absorbs the stress informally rather than triggering the formal resolution criterion. The biological core of the call survives all three attacks; the specific dated call does not survive them cleanly.

Why we are making the call We promote because the biological mechanism is real, the needle (donated human plasma litres) is genuinely inelastic and non-substitutable for the primary-immunodeficiency base, demand compounds from three independent curves, and the call is directionally correct over a 9-

year horizon. We downgrade the dated-call probability materially below our conviction in the thesis to reflect the three active kill paths: FcRn substitution in neurology, EU self-sufficiency policy, and the fragility of the formal-rationing resolution criterion. The structural foresight is sound; the exact dated call as written carries a live 60 percent probability of failing on its specific terms even as the underlying constraint tightens.

P5 · Donor-derived human platelet lysate is the hidden, non-synthesizable feedstock that becomes the binding constraint on industrial-scale allogeneic cell therapy as the field migrates off FBS onto a supplement that can only be made from human blood donors -- a donor pool that is already chronically short for clinical transfusion.

The boom: The allogeneic off-the-shelf cell therapy and regenerative-medicine wave: MSC products, iPSC-derived therapies, allogeneic CAR-T and NK products, and the broader stem-cell pipeline that has to expand cells to billions of doses in bioreactors. Regulatory and quality pressure is at the same time forcing the field off FBS, which is itself supply-capped and carries zoonotic risk, onto a human-derived substitute whose supply is biologically constrained and whose donor pool is already strained for ordinary clinical transfusion. · Domain: biotech and human health (cell/gene therapy upstream inputs, donor-derived biologics)

STRUCTURAL CASE

72%

OUR CALL, DATED

38%

RESOLVES

2033-12-31

Cells in culture need a serum supplement to proliferate. FBS is supply-capped and carries prion/viral/zoonotic risk that regulators increasingly reject. The leading xeno-free replacement, hPL, is made by pooling and lysing human platelet concentrates from volunteer donors. There is no synthetic route: the growth-factor cargo (PDGF, TGF-beta, IGF, EGF, FGF variants) is biologically inseparable from the human platelet, and pooling 40-80 donor units per production batch is required to average donor-to-donor variability. This hard-couples the cell-therapy industry's input supply to the human blood-donation pool. That pool is physically inelastic: it scales only with willing donors and apheresis-chair capacity, already runs seasonal and structural shortages for clinical transfusion, and has a hard biological ceiling. As allogeneic programs scale from thousands to millions of doses in liter-to-hundreds-of-liter bioreactors at 5% supplement, hPL demand competes with hospital transfusion services for the same platelet units. Recombinant growth-factor cocktails and chemically defined media exist but have not achieved broad GMP validation across MSC, iPSC-derived, and allogeneic NK cell types; regulatory requalification for process changes adds 2-5 years per product. Substitution is real but incomplete, and allogeneic scale-up is outrunning defined-media validation today.

WHY IT IS NOT PRICED YET

Market-research reports (Technavio, Polaris, TowardsHealthcare) name the hPL market and project 15% CAGR growth, so the category is tracked. The framing throughout, though, is growth-opportunity-in-a-niche, not supply-constraint-story. No equity analyst covers a public hPL pure-play. No forward price curve exists for hPL. No blood-bank or regulatory body has yet issued allocation guidance balancing transfusion versus manufacturing demand. The field frames hPL as the better, safer FBS replacement, not as a donor-derived bottleneck that cannot scale past human donation. The donor-platelet-diversion framing is absent from regulatory filings and blood-establishment policy. We see that as a genuine framing gap: the constraint is not yet priced at the level that matters.

WHERE THE PRICE SITS TODAY

No spot-price index for hPL exists. Catalog list prices from STEMCELL Technologies and MP Biomedicals are publicly visible but do not reflect GMP commercial-scale contract pricing, which is negotiated privately. No sell-side coverage of a hPL pure-play. Market-report coverage names the sector without pricing the constraint. The absence of a liquid price channel is consistent with the call being mispriced, not evidence it is already priced.

THE BINDING CONSTRAINT

GMP-grade human platelet lysate derived from pooled volunteer-donor blood-bank platelet units. The inelastic input is human-donor platelet mass itself (apheresis and whole-blood-derived platelets), which is biologically capped, already short for clinical transfusion, and cannot be synthesized or farmed.

WHAT WE ARE WATCHING

Tracked annually from 2026: (1) Estimated liters of GMP hPL consumed by cell-therapy and regenerative-medicine manufacturing versus total national platelet-unit collection, and the share of collected or expired platelet units diverted from transfusion into hPL production. (2) Real price per liter of GMP, pathogen-reduced, fibrinogen-depleted hPL and its trend, plus the count of cell-therapy sponsors disclosing hPL or human-platelet supply as a named supply-chain risk or qualifying second sources in regulatory filings or investor disclosures. (3) Whether blood establishments or regulatory bodies issue allocation guidance balancing transfusion versus manufacturing demand for platelets. Resolution requires real-terms hPL price rise plus at least three clinical-stage or commercial allogeneic programs publicly citing human-platelet or hPL supply as a rate-limiting or dual-sourced input.

WHAT WOULD PROVE US WRONG

Kill if, by 2033, a chemically defined or recombinant serum-free medium achieves broad GMP adoption across MSC, iPSC-derived, and allogeneic NK processes such that hPL is no longer the dominant supplement (major new allogeneic approvals run on defined media and hPL demand growth flattens). Also kill if scalable non-donor sources of the platelet growth-factor cargo emerge at cost-competitive GMP scale (iPSC-derived platelets, bioreactor-grown megakaryocytes, or recombinant factor cocktails), decoupling cell-therapy input supply from the human blood-donor pool. Also kill if hPL real price stays flat and no allogeneic sponsor cites human-platelet supply as a constraint through 2033.

HOW WE TRIED TO BREAK IT

Three attacks. First: the category is already in market reports, so it is priced. This weakens on inspection, because the reports price the growth opportunity in hPL sales, not the scarcity of donor platelet mass as a bottleneck for cell-therapy producers. Those are different claims pointing in opposite directions commercially. Second: defined media and recombinant supplements will substitute before the constraint bites. This is non-trivial and the most serious attack. Miltenyi TexMACS and others are active, and some MSC runs are already hPL-free. But no single defined medium matches hPL performance across the full cell-type space today, regulatory requalification adds years per product, and allogeneic scale-up is moving faster than defined-media validation. The kill criterion captures this exit. Third: the volume math does not close by 2033. Possible. Commercial allogeneic products at million-dose scale are still 5-10 years from dominance, and the diversion math may not become visible until post-2033. This attack does not kill the structural case, but it does compress the dated-call probability. The call survives all three with reduced but positive conviction. We hold the dated-call probability at 0.38 because the resolution bar (public price rise plus three named-input

disclosures) is specific, the timeline is tight relative to allogeneic scale-up, and defined-media progress is real.

Why we are making the call We promote because the biological mechanism is genuine and non-synthesizable, the inelastic needle is precisely identified (donor platelet mass), the field is systematically mispricing the dependency by framing hPL as a quality upgrade rather than a constrained input, and no liquid price channel or equity coverage currently reflects the donor-diversion risk. The kill line is honest and testable. We hold the dated-call probability to 0.38, not inflated, because defined-media substitution is a real competing path and the resolution bar requires public evidentiary disclosure that sponsors have structural incentives to avoid.

P6 · GLP-1 mass adoption structurally degrades the deceased-donor organ supply: by 2033 the binding constraint in solid-organ transplantation becomes the vanishing pool of young, low-comorbidity SCD organs, forcing value to migrate onto organ reconditioning (normothermic machine perfusion) and xeno/engineered organ infrastructure rather than donor recruitment.

The boom: The loud GLP-1 narrative is about supply and demand in obesity/diabetes therapeutics and downstream food/device/CDMO effects. The transplant-adjacent GLP-1 conversation is entirely about GLP-1 as therapy for recipients and for reversing donor-organ steatosis. Everyone treats GLP-1 as straightforwardly good for organ health. Nobody models GLP-1 as a structural drag on the donor pool itself. · Domain: biotech-health (transplantation / organ supply, GLP-1 second-order)

STRUCTURAL CASE

68%

OUR CALL, DATED

38%

RESOLVES

2033-12-31

The deceased-donor pool is a byproduct of how people die young and healthy. The overdose epidemic supplied roughly 42% of US deceased-donor growth between 2009 and 2019 because overdose victims are young with intact organs. That stream is collapsing: overdose-donor share fell from 16.7% in 2022 to 10.5% by Q1 2025 as overdose mortality dropped roughly 27%. Replacement donors are structurally worse, with post-inflection data showing median donor age up, diabetes +24%, hypertension +19%, and rising reliance on donation-after-circulatory-death (DCD) with measurably lower graft survival. GLP-1 mass adoption compounds this in a direction nobody models: it durably reduces obesity, diabetes, cardiovascular and cerebrovascular mortality at population scale, which are exactly the death mechanisms feeding the remaining marginal-donor stream, while doing nothing to replenish the young/traumatic-death SCD stream. Over a decade the deceased-donor pool gets older, smaller relative to demand, and increasingly composed of organs that need reconditioning to be usable. The forced response is normothermic machine perfusion to recondition marginal/DCD/steatotic organs, plus a structural pull toward xeno and engineered organs as the only elastic supply.

WHY IT IS NOT PRICED YET

The transplant field discusses the aging/comorbid donor-pool shift and the overdose-donor decline as separate phenomena, and separately celebrates GLP-1 as beneficial for organ health and recipients. The specific second-order synthesis (that GLP-1 mass adoption removes the metabolic/cardiovascular-death donor stream and thereby tightens the supply of usable young SCD organs, repricing reconditioning and engineered-organ infrastructure as the binding capacity) appears in zero published transplant or investor analysis. Sell-side covers TransMedics as a technology/logistics adoption story, not as a GLP-1 second-order consequence. The consensus has the sign of GLP-1's effect on transplantation backwards at the population-supply level. Note: the downstream asset (NMP/reconditioning equity) is already partially priced, since TMDX is a high-momentum public compounder, but the specific causal attribution via GLP-1 donor-pool degradation is genuinely absent from coverage, which makes this a structural-mechanism insight rather than an undiscovered equity trade.

WHERE THE PRICE SITS TODAY

TransMedics (TMDX) at \$605M 2025 revenue, 36% growth, active Buy-rated analyst coverage at \$109 price target, stock up 81% in 2025. NMP market reports project \$2B (2026) to \$6.5B+ (2033) at 18% CAGR. The reconditioning value migration is therefore partially priced as a technology-adoption story. The GLP-1 donor-pool-degradation causal path is not in any sell-side model or transplant analysis. Our read: the downstream consequence is partially priced on technology grounds; the specific mechanism named here is not priced. A structural call on the mechanism and its compounding is still mispriced; a simple "buy NMP stocks" trade is not.

THE BINDING CONSTRAINT

Standard-criteria-donor (SCD) organs: young, low-comorbidity deceased-donor solid organs usable without reconditioning. A pool fed by traumatic/overdose/cerebrovascular young death, now physically shrinking as those death mechanisms recede and GLP-1 removes the metabolic-death replacement stream.

WHAT WE ARE WATCHING

(a) SCD share of the US deceased-donor pool and median donor age (OPTN/SRTR annual data); (b) fraction of transplanted organs requiring machine perfusion or reconditioning before implant; (c) DCD and extended-criteria-donor share of total transplants; (d) installed normothermic-machine-perfusion device base and per-organ reconditioning revenue. Current anchors: overdose-donor share 16.7% to 10.5% (2022 to Q1 2025); TransMedics (TMDX) revenue \$605M in 2025, growing 36% year-over-year, active Buy coverage at \$109 target; NMP market projected \$2B in 2026 to \$6.5B by 2033 at 18% CAGR. GLP-1 contribution to donor-pool degradation not yet separated in any published donor-pool analysis.

WHAT WOULD PROVE US WRONG

By 2033, the deceased-donor SCD pool is stable or growing in absolute terms and median donor age is flat-to-falling (a new young-death source or expanded living/altruistic donation offsets the decline), AND machine-perfusion reconditioning has not become a standard step for a rising majority of transplanted organs. OR scaled xeno/engineered organs become the dominant marginal-supply solution, meaning SCD scarcity was bypassed rather than binding. OR GLP-1 population effects demonstrably fail to reduce the comorbid-death donor stream within the window (donor mortality mechanisms unchanged by 2030), falsifying the causal mechanism.

HOW WE TRIED TO BREAK IT

Three serious attacks survive review. First, the downstream asset is already priced: TMDX at high-multiple, high-momentum Buy-rated status means the market has partly re-rated NMP as the organ-quality solution. The "value migrates to reconditioning" call is thus partially already in prices, which weakens the investment edge. Second, the GLP-1 mortality timeline is slow: mass adoption began around 2023, and cardiovascular/metabolic death-rate effects at population scale take 5-10 years, so by 2033 GLP-1 may have only 10 years at scale and the donor-pool signal may not be separable from other variance within the window. Third, the kill criteria are plausible: DCD expansion is already accelerating as a partial offset, living donation is growing, and xeno-organ trials (FDA-authorized pig-to-human, 2024) are on a plausible timeline to reach limited commercial scale by 2030-2033, potentially bypassing rather than confirming the SCD-scarcity binding constraint. The structural direction is sound; the exact dated call (SCD scarcity becoming THE binding constraint and forcing observable value migration by 2033) is weaker because value migration is

already underway and the GLP-1 causal contribution is not separable within the window. The call survives as a structural insight but not at high precision on the exact resolution test.

Why we are making the call We promote on the structural mechanism: the GLP-1-as-donor-pool-degrader second-order insight is genuinely absent from transplant and investor analysis, the SCD organ needle is genuinely inelastic, and the directional trend (worse donor pool, rising NMP adoption) is already confirmed in real data. We hold the dated-call probability below 0.5 because the downstream equity is partially priced on technology grounds already, the GLP-1 causal contribution is not separable within the 2033 window, and alternative elastic responses (DCD expansion, xeno) may prevent SCD scarcity from becoming definitively "binding" in a measurable sense. This is a real structural insight worth tracking, not a high-confidence binary resolution.

Seeds considered and not promoted

These cleared the supply-side test but did not make the cut, usually because there was no clean way to express the trade or the move was already in the price. We list them because we would rather show our work than bury what we dropped.

SEED	PHYSICAL CASE	WHY NOT PROMOTED
By 2035, donor-rate ceiling forces formal indication-based IVIG rationing in at least one high-income system, with donor-collection-hours (not fractionation plant capacity) cited as the limiter.	Polyclonal immunoglobulin donor-rate ceiling / standing indication-based rationing	This is a near-duplicate of P4 (same plasma-donor needle, same FcRn/EU-self-sufficiency kill paths). P4 states the identical mechanism more strongly, with higher conviction in the thesis (0.82 vs 0.70) and a cleaner US-concentration metric, so we fold this one in rather than double-count the plasma thesis.
By 2034, purified Ra-226 legacy feedstock plus licensed Ac-227-free hot-cell separation capacity is the gating variable; at least three pivotal-stage programs publicly ration dose citing isotope feedstock or hot-cell limits.	Ra-226 feedstock + hot-cell separation capacity, three-program public-rationing test	This substantially overlaps P2 (Ra-226 finite legacy stock) and P1 (Ac-227-free purity) collapsed into one needle, and its resolution test is more fragile because it needs three separate companies to publicly disclose rationing, which sponsors are structurally incentivized to suppress. P1 and P2 already carry the two distinct physical needles more cleanly.
By 2032, Ra-226 target feedstock (not Ac-225 production capacity) becomes the explicitly cited gating variable for the radioligand pipeline.	Ra-226 as explicitly-cited gating variable by 2032	This is a direct near-duplicate of P2, with the same 2.5 kg legacy-stock needle and IAEA gleaning argument but a tighter, harder-to-meet resolution test (it requires explicit public Ra-226 citation by 2032 rather than by 2034). P2 captures the identical mechanism with more headroom on both timing and the price/inventory resolution channel.

Each call is dated, with the line that would prove us wrong fixed the day we make it. We do not move the goalposts: a call is superseded, never quietly edited, and graded in public when its date arrives.